RESEARCH & DESIGN OF UNMANNED AERIAL VEHICLES

UAV CONTEST 2027

AUTONOMOUS LOW-ALTITUDE LOGISTICS ECOSYSTEM

> From single-agent dynamic tracking to real-and-virtual coordinated multi-drone relay. Tackle real-world urban and warehouse logistics using the open **CABP Protocol** (CPAS Agent Beacon Protocol) in challenging GPS-denied environments.

COMPETITION DATES
SEPTEMBER 20 – 21, 2027

FLIGHT ARENA & VENUE
HO CHI MINH CITY, VIETNAM

COMPETITION FLIGHT TRACKS

INDOOR NETTED ARENA • 12M × 24M • UWB POSITIONING

TRACK 01 — SINGLE-UAV AUTONOMY

PRECISION LOGISTICS & LANDING

Autonomous obstacle navigation through low-altitude corridors with onboard real-time computer vision.

- **Phase 1:** Waypoint flight & static QR tag detection.
- **Phase 2:** Dynamic gate traversal & crosswind handling.
- **Grand Final:** Payload drop on moving Mobile Receiver (10 cm/s).

TRACK 02 • ADVANCED — 2-UAV SWARM COLLABORATION

COORDINATED DUAL-UAV RELAY

Multi-agent cooperative relay pipeline: *"Order Placement → Aerial Relay → Delivery"* via CABP.

- **Drone #1:** Autonomous pickup from automated depot pad.
- **Air Handshake:** Real-virtual peer beaconing via CABP.
- **Drone #2:** Mid-air handoff & autonomous end-user delivery.

HARDWARE REGULATIONS & OPEN-SOURCE STARTER KIT

AIRFRAME SIZE	MAX WEIGHT (MTOW)	POWER PACK	PAYLOAD CAPACITY
≤ 1.0 m	≤ 5.0 kg	≤ 6S LiPo	≥ 300 g
Motor-to-motor diagonal	Including payload & pack	Safe discharge rating	Standardized ball/capsule

FAST-DEPLOY OPEN-SOURCE STARTER KIT — 15-MIN ONBOARDING

Official SDK (ROS 2 / Python / C++) provided with pre-built Gazebo arena models, CABP sample scripts & trajectory planners.

SDK v4.0 READY

PARTICIPATION & TEAMS

- **Target:** Undergraduates, Master's & PhD engineering students worldwide.
- **Composition:** Maximum 6 student members per team + academic advisor(s).
- **Fields:** Robotics, Aerospace, Computer Vision, Embedded & AI.

CHAMPIONSHIP AWARDS & RECOGNITION

PRIZE POOL

GRAND CHAMPION	RUNNERS-UP	SPECIAL AWARD
GOLD TROPHY	1ST & 2ND PLACE	CABP SWARM
+ Tech Incubation	Cash Award & Medals	Best Multi-Agent Tech

JOIN THE AIRSPACE REVOLUTION

REGISTRATION OPEN

SCAN TO REGISTER

PARTNER TIER $5,000 USD (Co-Organizer, 1 Board Seat, 2 Teams) | Domestic: **2.5M VND** Intl: **$350 USD** ($500/2)

EARLY-BIRD REGISTRATION DEADLINE: **MAY 30, 2027**

SIMULATION PACKAGE SUBMISSION: **JULY 15, 2027**

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